Concise Midterm Defense Presentation Script (5-7 mins)

Slides 1-3: Introduction & Platform Vision

> "Good afternoon, respected professors and judges. My name is Yujie Feng, and my supervisor is Professor Jin Hou. Today, I'll present my midterm progress on our SaaS-oriented AI Inspection Platform. Our goal is to replace manual power grid inspections with a scalable AI backend that consumes multi-modal drone inputs—RGB, thermal, and telemetry. We are building a model registry supporting three core modules: Bird Nest, Oil Leak, and Infrared detection, all wrapped in a unified SaaS interface."

Slides 4-8: Module A (Bird Nest Detection)

> "Following a 'Vertical Slice First' strategy, my first milestone was Module A: Bird Nest Detection. I trained a Mask R-CNN model using mixed precision. To prevent overfitting on our limited 54-image dataset, I strictly used deterministic augmentations, focusing on a custom Random IoU crop rather than physically implausible generative AI. The results were exceptional: both bounding box and segmentation AP50 reached 1.0. This pixel-perfect instance segmentation is critical because it precisely outlines the hazard zone needed for dispatching safe work orders."

Slides 9-11: Module B (Oil Leak Detection)

> "For Module B, Oil Leaks, the priority is fast, lightweight alerting. I merged a State Grid VOC dataset with a CSDN YOLO dataset into a unified format and trained a YOLOv8-small model. Our initial tests yield a high precision of 0.83 but a conservative recall of 0.34. The model is highly accurate when it flags a leak, actively avoiding false-positive 'alert fatigue'. The immediate next step is expanding training on the fully merged dataset to boost recall."

Slides 12-13: Module C (Infrared Detection)

> "Module C handles Infrared thermal imaging. Instead of a heavy deep-learning model, I implemented a fast OCR and pixel-mapping hybrid approach. The algorithm reads the image's temperature legend via EasyOCR to establish a linear pixel-to-temperature map. It then thresholds the grayscale image based on a user-defined alarm trigger. To ensure robustness, I engineered two strict noise masks that automatically suppress the surrounding scale bars and white factory watermarks, virtually eliminating false positives."

Slides 14-15: SaaS Architecture & Demo

> "All three models are seamlessly orchestrated by a single Flask Microservice. **[Action: Point to diagram on Slide 14]** As shown, the API Gateway actively routes batch-uploaded images to the correct inference engine based on user requirements. The backend dynamically fetches weights from the model registry and bundles the bounding boxes and visual overlays into a JSON response. We've already wrapped this into a functional Web UI demonstrating multi-tenant batch inference and history tracking."

Slides 16-18: Challenges & Next Steps

> "While the functional prototype is complete, the main challenge ahead is moving from a 'demo' to an 'enterprise' state. Technically, we need to handle cross-scene

> generalization to prevent performance drops in real-world haze. Architecturally, we need to upgrade our Flask deployment with asynchronous task queues to support high-concurrency drone streams. In the next 10 weeks, I will focus on refining the multi-tenant SaaS backend, retraining the oil leak module, and conducting end-to-end multi-user testing.
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> Thank you. I welcome your questions and advice."